Sunghwan Bae

MS student
Gwangju Institute of Science and Technology (GIST)
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EDUCATION

Mar. 2025 ~ Present	**Gwangju Institute of Science and Technology** School of Mechanical Engineering *Master Student*	Gwangju, Korea
Mar. 2019 ~ Feb. 2025	**Gwangju Institute of Science and Technology** School of Mechanical Engineering Thesis: Improving Nonnegative Matrix Factorization for Muscle Synergy Analysis :Minimizing Synergy Entropy *B.S student* GPA: 3.74 / 4.5	Gwangju, Korea

RESEARCH INTERESTS

- Biomechanics
- EMG - Electromyography
- Human–robot interaction

RESEARCH EXPERIENCES

- **2023 - Present, Muscle Synergy Analysis**
  - Improving the Non-Sparsity problem of Non-negative Matrix Factoriztion for muscle synergy
  - Assessment of neural control during gait using muscle synergy analysis
- **2024 - Present, Free-Energy-Principle-Based Knee Assist Robot**
  - Designed and configured motion-capture, force-plate, and EMG sensor setups for gait data acquisition
  - Muscle activation data analysis.
- **2024, Table Tennis Robot**
  - Developed a predictive ball-trajectory model and optimized striking points
  - Robot posture to deliver the ball to desired location

CONFERENCES

1. Sunghwan Bae, Pilwon Hur, "Improving the Non-Sparsity problem of Non-negative Matrix Factoriztion for muscle synergy-based Gait Intention Estimation", ICROS, Jeonju, Korea (Jun. 2025) - Poster
2. Sunghwan Bae, Pilwon Hur, "Improving Nonnegative Matrix Factorization for Muscle Synergy Analysis: Minimizing Synergy Entropy", ICROS, Daejeon, Korea (Jul. 2024) - Poster

TEACHING & ASSISTANT EXPERIENCE

- **2025 Spring**, Signal & Systems in Mechanical Eng. - Teaching Assistant
- **2024 Winter**, Wearable Robotics Program for High School Students - Assistant
- **2023 Spring**, General Physics - Teaching Assistant

AWARDS AND HONORS

- Best Paper Award (President's Award), Department of Mechanical Engineering, Bachelor's Program , Gwangju Institute of Science and Technology, Korea (Feb. 2025)
- GIST Creative Convergence Competition, Table Tennis Robot Contest (3rd Place), Gwangju Institute of Science and Technology, Korea (Aug. 2024)
- Academic scholar ship, Gwangju Institute of Science and Technology, Korea (Dec. 2020)

SKILLS AND TECHNIQUES

- Programming Languages: Julia, Python, R, C++, C
- Software : Visual3D
- Experimental Equipment:
  - Nokov Motion-Capture Systems
  - Delsys Trigno EMG
  - AMTI Force Platforms
- Data Analysis & Methods:
  - Biomechanical Analysis (Inverse Dynamics, EMG analysis)
  - Muscle synergy analysis
  - Numerical optimization & Simulation